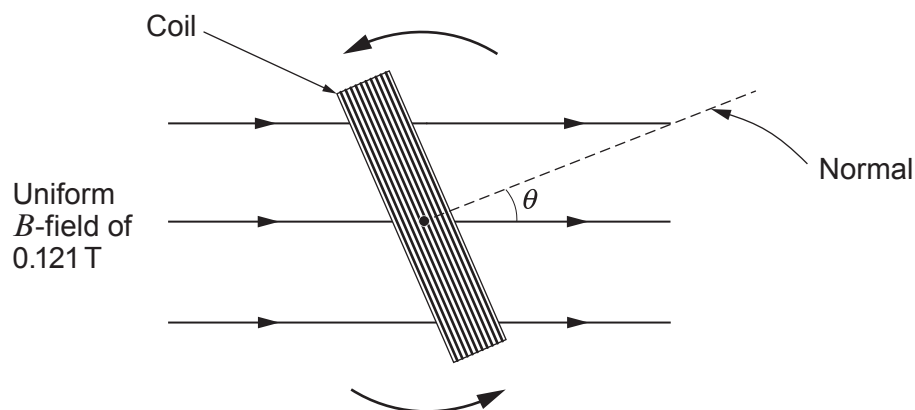


- (b) A rectangular coil rotates at a constant angular velocity within a uniform magnetic field of 0.121 T. The coil has 70 turns and cross-sectional area 59 cm². The diagram below shows the coil, looking along the axis of rotation.



- (i) Calculate the flux linkage of the coil when $\theta = 23^\circ$.

[2]

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- (ii) As the coil rotates, explain what values of θ provide the maximum **and** minimum values of the induced emf.

[2]

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